KRITIS-PH

Model Documentation

Version 2.0 — Internal Technical Reference

Status	Internal Draft
Scope	Philippines Transmission Grid + Tropical Cyclone Risk
Model	Screening-level, deterministic (v5r baseline)
Revised	April 13, 2026

Note: Model results are conditional on assumptions, parameters, and structure. They are not absolute truths.

Document Changelog

Version	Date	Author	Changes
1.0	—	—	Initial skeleton: scope, assumptions, parameters, validation philosophy
2.0	April 13, 2026	—	Layout overhaul; section expansions; notation table; callout boxes; TODO markers; research agenda detail

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1 Executive Summary

KRITIS-PH is a **screening-level power grid disruption model** for the Philippines. It quantifies the approximate number of people whose electricity supply is put at risk when a tropical cyclone crosses the archipelago. It does *not* aim to replace engineering damage assessments or precise outage simulations.

What the model does in one sentence:

Core Function

Given a storm track and intensity record, the model identifies which transmission lines likely fail, how the resulting network fragmentation degrades service, and how many people are exposed to degraded or lost electricity supply.

Primary use cases:

- **Event ranking** — compare historical typhoons by estimated grid impact
- **Vulnerability mapping** — identify structurally exposed nodes and corridors
- **Decision-support prototyping** — rapid pre-event or scenario-based assessments

Non-use cases:

- Engineering-grade damage estimation
- Exact outage durations or restoration timelines
- Economic loss quantification

Interpretation Boundary

All model outputs are *relative and conditional*. A higher impact index means a storm is estimated to be *comparatively more disruptive*—it does not constitute a verified damage figure.

2 Model Overview

2.1 Conceptual Pipeline

The model follows a five-stage causal chain:

$$\textbf{Hazard} \xrightarrow{\text{threshold}} \textbf{Line Failure} \xrightarrow{\text{topology}} \textbf{Fragmentation} \xrightarrow{\text{adequacy}} \textbf{Service Degradation} \xrightarrow{\text{proximity}} \textbf{Affected Population}$$

Each arrow represents a modelling choice with embedded assumptions. The stages are described in detail in section 3.

2.2 Model Classification

Attribute	Description
Deterministic (baseline)	No random draws; canonical parameters produce a single output
Threshold-based	Line failure triggered by exceedance of a scalar hazard intensity
Graph-driven	Network effects propagated through adjacency/connectivity analysis
Proxy-based	Service adequacy approximated via capacity-to-population ratio
Screening-level	Optimised for comparative insight, not engineering precision

2.3 Spatial Domain

The model is restricted to the Philippines national bounding box:

- Latitude: 4.0N – 22.0N
- Longitude: 116.0E – 128.0E

Storms are included if and only if their track intersects this box.

3 Formal Model Summary

3.1 Notation Reference

Symbol	Type	Meaning
$G = (B, L)$	Graph	Power network: buses (nodes) B, transmission lines (edges) L
G_t	Graph	Residual network at time t after line failures
T	Set	Discrete time steps along storm track
S	Set	Storm event catalogue
x	Point	Geographic location (lon, lat)
$x_{t'}$	Point	Storm centre position at time step t'
$V_{t'}$	Scalar	Maximum sustained wind speed at time t'
$V_{\max}$	Scalar	Normalisation reference wind speed
$H(x, t)$	Scalar	Cumulative normalised hazard at location x by time t
$H_l(t)$	Scalar	Maximum hazard experienced by line l by time t
θ	Scalar	Failure threshold (canonical: `0.55`)
$f(d)$	Function	Spatial decay function
$d(\cdot)$	Function	Euclidean (or great-circle) distance
$R_{\max}$	Scalar	Radius of maximum winds (canonical: `35 km`)
α	Scalar	Decay exponent (canonical: `1.35`)
c_{inner}	Scalar	Inner-core hazard floor (canonical: `0.85`)
$L_{\text{failed}}(t)$	Set	Cumulative set of failed lines up to time t
C_{total}	Scalar	Total generation capacity in a connected component
P_{total}	Scalar	Total population served by that component
$b(p)$	Function	Nearest-bus assignment for population point p
I_{grid}	Scalar	Composite grid impact index
w_1, w_2, w_3	Scalars	Impact index weights

3.2 System Definition

The power network is modelled as an undirected graph:

$$G = (B,\, L) \tag{1}$$

where B is the set of buses (substations/nodes) and $L \subseteq B \times B$ is the set of transmission lines.

3.3 Hazard Model

The normalised hazard at any location x at time t is defined as the running maximum over all preceding track positions:

$$H(x,t) = \max_{t' \le t} \left[\frac{V_{t'}}{V_{\max}} \cdot f(d(x,\, x_{t'})) \right] \tag{2}$$

This construction ensures that hazard *accumulates monotonically* — once a location has been exposed to a hazard level, that level cannot decrease. This reflects the assumption that structural damage does not self-repair during an event (section 4.2).

3.4 Spatial Decay Function

$$f(d) = \begin{cases} c_{\text{inner}} & d \le R_{\max} \\ \left(\dfrac{R_{\max}}{d} \right)^{\alpha} & d > R_{\max} \end{cases} \tag{3}$$

The inner-core floor c_{inner} prevents the hazard from collapsing to zero near the eye of the storm (where wind speeds also dip). Outside $R_{\max}$, hazard falls off as a power law with exponent α.

Radial Symmetry

The decay function is symmetric in d. Asymmetric wind structures (e.g. quadrant-dependent intensification) are not modelled. This is a known limitation; see section 7.

3.5 Line Failure

Each transmission line l is represented by a set of k sample points (canonical: 7). The line-level hazard is the maximum over those points:

$$H_l(t) = \max_k H(x_k,\, t) \tag{4}$$

A line fails if its hazard exceeds the failure threshold:

$$l \in L_{\text{failed}}(t) \iff H_l(t) > \theta \tag{5}$$

3.6 Cumulative Failure Set

Failures accumulate over time; no repair occurs within the event window:

$$L_{\text{failed}}(t) = \bigcup_{t' \le t} \{ l \in L : H_l(t') > \theta \} \tag{6}$$

3.7 Residual Network State

The network at time t is the original graph with all failed lines removed:

$$G_t = \big(B,\ L \setminus L_{\text{failed}}(t)\big) \tag{7}$$

Connected components of G_t are computed after each failure step. Buses that are no longer connected to any generator constitute *unserved* buses.

3.8 Service Adequacy

For each connected component C of G_t:

$$\text{Adequacy}(C) = \frac{C_{\text{total}}}{P_{\text{total}}} \tag{8}$$

where C_{total} is the total installed generation capacity (MW) and P_{total} is the total population assigned to buses in C. Two thresholds are used:

Mode	Meaning	Canonical threshold
Multi-bus component	Adequacy below which the component is deemed *undersupplied*	`0.35`
Single-bus island	Adequacy below which a lone bus is deemed *unserved*	`0.75`

Proxy Rationale

Using MW/capita as a service proxy avoids the need for full power flow simulation. It captures the intuition that a component with very little generation relative to population will face load-shedding or blackout conditions. However, it does not model transmission constraints, voltage stability, or dispatch logic.

3.9 Population Assignment

Each population raster cell p is assigned to the geographically closest bus:

$$b(p) = \arg\min_{b \in B} d(p,\, b) \tag{9}$$

Population is static (WorldPop 2020, 1 km resolution) and does not evolve over the simulation.

3.10 Grid Impact Index

The composite impact index aggregates three normalised quantities:

$$I_{\text{grid}} = w_1 \hat{P}_{\text{affected}} + w_2 \hat{N}_{\text{failed}} + w_3 \hat{N}_{\text{unserved}} \tag{10}$$

Component	Weight	Description
$\hat{P}_{\text{affected}}$	w_1	Population in degraded or unserved components (normalised)
$\hat{N}_{\text{failed}}$	w_2	Number of failed lines (normalised)
$\hat{N}_{\text{unserved}}$	w_3	Number of unserved buses (normalised)

Weight Values and Rationale

The canonical weights are $w_1 = 0.60$, $w_2 = 0.25$, $w_3 = 0.15$, reflecting a deliberate population-first priority:

- $w_1 = 0.60$ **(affected population proxy)** — given primacy as the most direct expression of human exposure to service disruption; a storm that fragments the grid without affecting populated areas is treated as less critical than one affecting densely populated nodes.
- $w_2 = 0.25$ **(failed lines)** — captures structural grid damage; weighted second as a proxy for infrastructure stress that is independent of population distribution.
- $w_3 = 0.15$ **(unserved buses)** — captures topological isolation; weighted least because unserved bus count is partly co-determined by line failure count and adequacy thresholds already captured by w_1.

These weights were set by *exploratory judgment* during model development, not through formal expert elicitation or empirical calibration. Their sensitivity should be assessed as part of Phase 1 robustness work (section 11). The index is min-max normalised within the storm catalogue before weighting, so weights are relative importance scores rather than absolute scaling factors.

4 Assumptions

Understanding the assumption stack is more important than reading the outputs. Every assumption narrows the interpretation boundary.

4.1 Hazard Assumptions

Wind-only Only wind is modelled. Storm surge, flooding, and rainfall-induced failures are excluded. This is the single largest source of systematic underestimation for landfalling typhoons with large rainfall footprints (e.g. Haiyan 2013, Ulysses 2020).

Radial symmetry Hazard decays uniformly with distance from the storm centre. Observed asymmetries (right-of-track intensification in the Northern Hemisphere due to storm translation speed) are ignored.

Max-over-time Damage accumulates monotonically (eq. (6)). No fatigue model, no progressive weakening. A line that survives one near-miss is treated identically to one that has never been exposed.

Normalisation Hazard is expressed as a fraction of $V_{\max}$, making the index relative rather than absolute. Results are not comparable across different $V_{\max}$ choices.

4.2 Infrastructure Assumptions

Binary failure Lines are either fully functional or completely failed. Partial damage, reduced

capacity, or de-rating are not represented.

Uniform vulnerability
All lines share the same failure threshold θ. In reality, line vulnerability varies with tower type, age, terrain, conductor type, and maintenance history.

No repair
Once failed, a line remains failed for the duration of the simulation. Post-storm restoration dynamics are outside the model scope (see Research Agenda, section 11).

No overload cascades
Failure is hazard-driven only. Overload-triggered cascading failures (common in real grid disturbances) are not modelled.

4.3 Network Assumptions

Topology-only
Connectivity is the sole network metric. No power flow equations, no Kirchhoff constraints, no voltage or frequency dynamics.

Instantaneous cascade
Connectivity changes propagate immediately when a line fails. No propagation delay or protection relay behaviour is modelled.

4.4 Service & Exposure Assumptions

Capacity proxy
Generation capacity per capita proxies service adequacy. This ignores demand-side dynamics, import/export across interconnectors, and dispatch optimisation.

Binary service state
A bus (and all population assigned to it) is either *served* or *unserved*. Partial supply (e.g. rolling blackouts) is not represented.

Nearest-bus assignment
Population cells are assigned to the closest bus by straight-line distance. Grid connectivity, road networks, or actual distribution service zones are not used.

Static population
WorldPop 2020 figures are fixed. Pre-event evacuation, seasonal variation, and demographic change are not accounted for.

5 Canonical Parameters

The following parameter set defines the *canonical v5r* model configuration. All comparative results in the event ranking are produced with these values.

Parameter	Symbol / Key	Description	Value
Failure threshold	θ	Normalised hazard level triggering line failure	`0.55`
Radius of max winds	$R_{\max}$	Inner-core radius in km	`35 km`
Decay exponent	α	Power-law decay rate beyond $R_{\max}$	`1.35`
Inner-core floor	c_{inner}	Minimum hazard within $R_{\max}$	`0.85`
Line sample points	—	Points per line used to evaluate H_l	`7`
Search radius	—	Maximum distance for track–node association	`200 km`
Minimum wind	—	Track records below this speed are ignored	`30 km/h`
Min. total generation	—	Components below this are treated as non-generating	`1 MW`
Adequacy threshold (multi)	—	Component adequacy threshold (multi-bus)	`0.35`
Adequacy threshold (single)	—	Component adequacy threshold (single-bus island)	`0.75`
Random seed	—	For any stochastic sub-routines	`42`

Threshold Sensitivity

The failure threshold θ is the single most influential parameter in the model. A shift from 0.55 to 0.45 can substantially increase the number of failed lines and affected population. Always perform a θ-sweep before drawing cross-event comparisons.

Parameter Provenance

The canonical parameter values were set through *exploratory judgment* during iterative model development, not through formal calibration, expert elicitation, or fitting to observational data. Specifically:

- $\theta = 0.55$: chosen as a mid-range threshold that produces a plausible number of failures for strong typhoons without triggering failures for weak peripheral events. No empirical fragility data was used.
- $R_{\max} = 35\,\mathbf{km}$: representative of a moderate tropical cyclone radius of maximum winds; treated as a fixed characteristic scale rather than a storm-specific estimate.
- $\alpha = 1.35$: selected to produce a gradual outer decay that ensures storms with large footprints affect transmission corridors at distance; value is in the range of stylized wind-decay exponents found in simplified hazard models but is not fitted to Philippine cyclone observations.
- $c_{\mathbf{inner}} = 0.85$: prevents the inner-core hazard from collapsing to zero near the eye; value is heuristic and chosen to maintain elevated hazard across the inner-core zone.
- **Adequacy thresholds (0.35 / 0.75)**: set by inspection of baseline adequacy distributions across the PyPSA-PH network; intended to distinguish clearly undersupplied components from plausibly served ones.

A structured parameter justification exercise — including comparison to existing Philippines cyclone wind-field literature and sensitivity analysis — is identified as a Phase 1 research priority (section 11).

6 Sensitivity Framework

6.1 Purpose

Sensitivity analysis serves two roles: (1) understanding which parameters drive model output, and (2) bounding the uncertainty in results given parameter uncertainty.

6.2 Key Parameters by Sensitivity Tier

Tier	Parameter	Expected impact
High	Failure threshold θ	Dominates failure count and affected population
High	Adequacy definition	Directly controls served/unserved classification
High	Network topology (data)	Structural changes cascade through all downstream metrics
Medium	Decay exponent α	Controls spatial extent of failures
Medium	$R_{\max}$	Controls inner-core hazard footprint
Low	Line sample points	Computational precision; low impact above 5 points
Low	Search radius	Edge effect only for weakly-affecting distant storms

6.3 Sensitivity Analysis Types

Local Small perturbations (±10%) around canonical values. Tests whether the model is near a bifurcation (e.g. threshold crossings).

Scenario Alternative named configurations (e.g. `low-threshold`, `high-decay`). Enables bracketing of plausible outcomes.

Structural Changes to model architecture (e.g. probabilistic vs. deterministic failure, Voronoi vs. nearest-bus assignment). Tests robustness of conclusions to design choices.

Key Sensitivity Finding

Model behaviour near the failure threshold is highly nonlinear. Because lines fail discretely, small changes in θ near the median hazard level of a storm can produce large jumps in N_{failed}. Rank-order stability should be verified across a θ-sweep, not just at the canonical value.

7 Limitations

Read Before Interpreting Results

The following limitations are not edge cases—they are fundamental to the model architecture. Each one constitutes a known gap between model behaviour and real-world system behaviour.

1. **Wind-only hazard.** Flooding and rainfall, often the dominant damage mechanisms for slow-moving or recurving typhoons, are entirely absent.
2. **Simplified wind field.** No asymmetry, no boundary layer effects, no terrain interaction.
3. **Binary failure.** Real infrastructure degrades continuously; the threshold model cannot represent partial damage.
4. **Uniform vulnerability.** Line age, construction standard, and local exposure are not differentiated.
5. **No repair dynamics.** Restoration timelines, crew availability, and grid restoration sequencing are out of scope.
6. **No power flow physics.** Thermal limits, voltage collapse, and frequency instability are not modelled.
7. **Binary service classification.** Households experiencing rolling blackouts are treated identically to fully-served households.
8. **Stylised population assignment.** Service zones do not follow actual distribution network topology.
9. **No socio-economic differentiation.** Critical facilities (hospitals, water plants), economic activity, and social vulnerability are not weighted.
10. **Proxy mismatch in validation.** EM-DAT figures capture reported affected persons, which depend on reporting practices and include multi-hazard effects not in the model.
11. **Limited uncertainty quantification.** The canonical model is fully deterministic; parameter uncertainty is not propagated to output distributions.

8 Data Sources and Preprocessing

8.1 Input Datasets

Dataset	Description	Access
IBTrACS	Tropical cyclone best-track data (via `tropycal`)	Public
PyPSA-PH	Philippines transmission network: buses, lines, generators	Internal
WorldPop 2020	Population raster, 1 km resolution	Public
EM-DAT	Disaster loss database (optional validation)	Registered

8.2 Preprocessing Pipeline

1. Filter IBTrACS tracks to Philippines bounding box; sort by time
2. Discard track records with wind speed below minimum threshold (30 km/h)
3. Sample each transmission line into 7 evenly-spaced points
4. Map WorldPop raster cells to nearest PyPSA-PH bus
5. Aggregate generator capacities per bus
6. Compute baseline adequacy for the intact network

Reproducibility

To ensure reproducibility, the following version information should be recorded at each model run. A `data_manifest.yaml` in the project root is the recommended format.

- **IBTrACS**: record the download date and version string returned by `tropycal` (e.g. `v04r00`); the archive is updated periodically and historical tracks can be revised.
- **PyPSA-PH**: record the commit hash or release tag of the network data repository; topology changes directly affect all model outputs.
- **WorldPop 2020**: the 1 km constrained individual countries dataset; this is a static file but the download URL and file hash should be recorded.
- **EM-DAT**: record the download date and query filters (country: Philippines, disaster type: Tropical Cyclone, year range); EM-DAT revises historical records and figures may change between downloads.

9 Execution and Outputs

9.1 Workflow

1. Data preprocessing (§ 8)
2. Baseline network construction
3. Per-storm simulation: hazard field → line failure → network state
4. Network fragmentation analysis (connected components)
5. Service adequacy evaluation per component
6. Impact aggregation: compute I_{grid} per storm (§ 3.10)
7. Event ranking

8. Validation and robustness analysis (§ 10)

9.2 Output Artefacts

Output	Description
Event ranking table	Storms sorted by I_{grid}; columns include $\hat{P}_{\text{affected}}$, $\hat{N}_{\text{failed}}$, $\hat{N}_{\text{unserved}}$
Storm time series	I_{grid}, failed-line count, affected population over storm duration
Line failure hotspot map	Spatial density of line failures aggregated across events
Bus disruption map	Per-bus frequency and severity of service degradation
Flagship storm diagnostics	Detailed trace for selected high-impact events
Validation statistics	Spearman rank correlation, Top-K overlap, scatter plots

9.3 Output Interpretation Rules

How to Read Model Outputs

- All values are **relative**, not absolute—compare storms *within* the model, not against external benchmarks without caution.
- Outputs are **conditional on the canonical parameter set**—a different θ will produce a different ranking.
- Use outputs as **structural indicators**: which corridors fail most often? Which island groups are most isolated? These structural signals are more robust than point estimates.

10 Validation Philosophy

10.1 Core Principle

Validation Objective

Validation tests *structural plausibility*, not numerical accuracy. The question is not "does the model reproduce EM-DAT figures?" but rather "does the model rank storms in a structurally sensible order?"

10.2 Validation Metrics

Metric	Description	Interpretation
Spearman ρ	Rank correlation between model I_{grid} and EM-DAT affected persons	Moderate positive expected
Top-K overlap	Overlap fraction between model top-K and observed top-K events	>50% is plausible
Scatter diagnostics	Visual inspection of model vs. observed pairs; outlier identification	Outliers are informative

10.3 Interpreting Validation Results

Moderate correlation is the expected—and *correct*—outcome, because:

- The model captures only one component of the real impact system (wind → grid)
- Real-world losses include rainfall, surge, response capacity, and reporting artefacts
- EM-DAT figures have their own uncertainty and multi-hazard aggregation

Suspicious Outcomes

Perfect correlation suggests overfitting to the validation dataset or data leakage—not model quality.

Absence of outliers suggests the validation proxy is uninformative, or that the model is interpolating rather than capturing causal structure.

Validation Scope and Future Extensions

The current implementation uses EM-DAT as the sole external reference. EM-DAT is a broad disaster impact database — it reflects total reported affected persons across all hazard pathways, not grid-specific outage severity. This creates a structural proxy mismatch: the model ranks storms by grid disruption, while EM-DAT reports multi-hazard societal impact.

The validation therefore targets *ordinal plausibility* rather than numerical agreement. Three checks are implemented:

- **Top-K retrieval**: do historically recognized major storms (e.g. Haiyan 2013, Ompong 2018, Ulysses 2020) appear disproportionately in the model's upper tail?
- **Spearman rank correlation**: does the model's I_{grid} ordering show positive rank correlation with EM-DAT affected-population figures for matched events?
- **Scatter diagnostics**: are outliers interpretable (e.g. rainfall-dominant storms ranked lower than their EM-DAT severity would suggest)?

Candidate additional validation sources for future work include NGCP (National Grid Corporation of the Philippines) post-event line outage reports and VIIRS nighttime light satellite imagery as an independent proxy for service-area blackout extent. Both are identified as Phase 1 priorities (section 11).

11 Research Agenda

Development is organised in four phases, from near-term model improvement to long-horizon system extensions.

Phase 1 — Model Robustness `[near-term]`

- Probabilistic failure models: replace binary threshold with fragility curves (e.g. lognormal)
- Parameter ensemble analysis: Monte Carlo or Sobol sensitivity over $(\theta, \alpha, R_{\max})$
- Expanded validation metrics: nighttime lights, NGCP outage reports

Phase 2 — Multi-Hazard Extension [medium-term]

- Multi-hazard integration: rainfall and flooding as separate failure triggers
- Heterogeneous vulnerability: line-specific fragility based on construction type and age
- Improved exposure mapping: census-block level population; critical infrastructure weighting

Phase 3 — System Dynamics [medium-term]

- Recovery and repair dynamics: time-to-repair functions per line type; crew deployment model
- Flow-based power system modelling: DC power flow or linearised OPF
- Dynamic cascading failures: overload-triggered secondary failures

Phase 4 — Integration [long-term]

- Coupling to financial risk models: expected annual loss (EAL) estimation
- Agent-based system extensions: household-level impact pathways
- Climate scenario integration: track-frequency and intensity changes under SSP scenarios

12 Interpretation Guidance

12.1 What the Model Can Say

- **Relative event severity:** Storm A is estimated to be more grid-disruptive than Storm B under the canonical assumptions.
- **Structural vulnerabilities:** Specific lines and nodes that appear across many high-impact events.
- **Network topology effects:** How the island geography and network structure amplify fragmentation.

12.2 What the Model Cannot Say

- Exact damage extents or damage costs
- Outage duration or restoration timeline
- Economic losses or welfare impacts
- Impacts from non-wind hazards (flooding, surge)

Core Rule

$$\text{assumptions} \times \text{parameters} \times \text{structure} \longrightarrow \text{model behaviour}$$

Changing any element of this triple changes the output. Results should always be reported with their conditioning assumptions stated explicitly.

13 Glossary

Term	Definition
Adequacy	Ratio of generation capacity to population in a connected component
Binary failure	A failure model in which components are either fully functional or completely failed
Bus	A node in the power network, typically a substation or generator connection point
Connected component	Maximal subgraph of G_t in which all buses are reachable from each other
Cumulative hazard	The running maximum of normalised wind hazard experienced at a location
Decay exponent	Parameter α controlling how fast hazard falls off with distance
Failure threshold	Scalar θ; hazard level above which a line is considered failed
Fragility curve	Probabilistic alternative to binary failure; gives $P(\text{fail} \mid \text{hazard})$
Screening model	Model designed for rapid comparative analysis rather than precision simulation
Unserved bus	A bus in G_t not connected to any generating bus

[TODO] Glossary Completion

Add definitions for: IBTrACS, PyPSA-PH, EM-DAT, $V_{\max}$ convention, track record, WorldPop methodology.

Meta Takeaway

Simple hazard rules interacting with network topology and exposure proxies generate complex, interpretable system behaviour.

Understanding assumptions is more important than interpreting outputs as reality.